

it transforms back to metallic iron as the battery is recharged. During discharging, Fe changes into ions on the anodic electrode, while oxygen transforms into hydroxide ions at the cathodic electrode. The diffusion

of oxygen into the battery occurs through the gas diffusion layer.

During the transition of Fe into metallic ions, electrons are produced. Metallic ions subsequently dissolve into the electrolyte. All these steps

are reversed during charging.

The major problems faced by the iron electrode with alkaline electrolytes are the hydrogen evolution reaction and the formation of insulating iron hydroxide during charge-discharge processes. Usually, while charging an iron electrode, water is decomposed to produce hydrogen, and consequently, an amount of the energy that was supposed to be stored in the battery is wasted as a result of parasitic hydrogen evolution.

Additives like Bi_2S_3 , K_2S , or aliphatic sulphides may be added to the iron electrode to increase the charging efficiency, increase the porosity of the iron powder, and decrease the effect of the passivating layer. Another approach to restrain these drawbacks is by adding ionic liquid (1-methyl-methylimidazolium L-(+)-lactate (EML) as an electrolyte additive to suppress both hydrogen release and passivation formation on the electrode surface during battery discharge by forming iron lactate instead of iron oxide. As a result, the Fe-air battery achieves

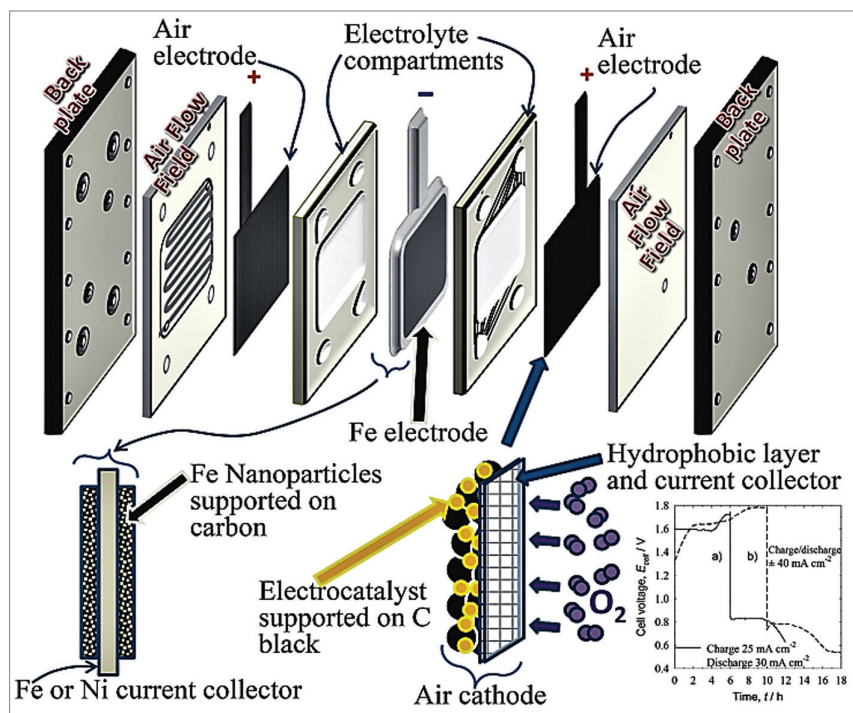


Fig. 8: Exploded view of iron-air battery

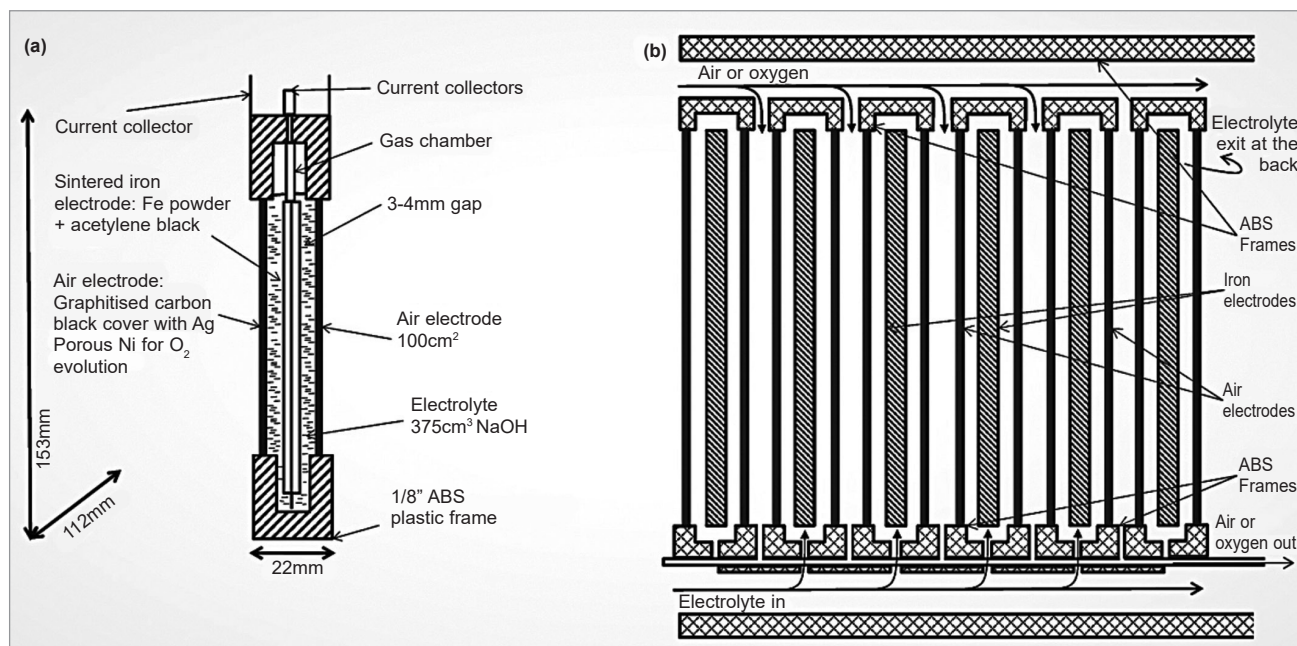


Fig. 9: Sectional view, (a) Single cell configuration, (b) Six cell configuration